

OEE: The TPM Scorecard

$$\text{OEE} = \text{Availability} \times \text{Performance} \times \text{Quality}$$

AVAILABILITY

$$\frac{\text{Run Time}}{\text{Planned Production Time}}$$

**Downtime
Losses**

PERFORMANCE

$$\frac{\text{Ideal Cycle Time} \times \text{Total Count}}{\text{Run Time}}$$

**Speed
Losses**

QUALITY

$$\frac{\text{Good Count}}{\text{Total Count}}$$

**Quality
Losses**